Corrected MCQs

1. **B.** $u_1 = 5$ then $u_2 = 8$. Responses 5 and 11 confuse ranks; 2 $^4 = 16$ does not correspond to the Additive relationship.
2. **C.** We set a row. n and we assume $P(n)$ to demonstrate $P(n+1)$. Check the first row is initialization, and directly assuming the conclusion would be circular.
3. **B.** The indices are 0, 1, 2, 3, 4 : five terms. The number of terms is last index minus first No more than one.
4. **B.** The sign of differences proves strict growth. $u_n = n - 100$ shows that neither positivity nor increase or convergence are not imposed.
5. **C.** The exit occurs when the condition $u \leq 10$ becomes false, so for $u > 10$. Reaching exactly 10 is not enough to stop the loop.
6. **B.** Monotonous convergence gives a maximum limit equal to 7. -1 refutes the limit 7 and positivity; 7 $-1/(n+1)$ is not geometric.
7. **A.** A value unchanged by the update verifies $L = 0{,}5L + 4$, or $L = 8$. This equation alone does not prove convergence.
8. **B.** $v_{n+1} = 0{,}5u_n + 4 - 8 = 0{,}5(u_n - 8) = 0{,}5v_n$. The subtraction of the fixed point removes the constant term.
9. **B.** The gap $|u_n - 6|$ measures the distance from u_n 6. Strict inequality $|u_n - 6| < 0{,}01$ exactly described Open interval requested.
10. **C.** A finished calculation informs about the terms actually calculated. A property at all ranks requires proof; convergence requires a study of the whole sequence.

True/false. False: $u_n = n$ True: 2 $^{-n} > 0$, therefore $6 - 2^{-n} < 6$. False: with $f(x) = -x$, point fixed is 0 but $u_0 = 1$ generates $(-1)^n$, which diverges.

Corrected applications and support

Application 1. $v_0 = 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. Items are: $(0;1), (1;3), (2;7), (3;15)$. $v_2 + 1 = 8$ but $v_{2+1} = 15$.

Application 2. Three updates: Repeat `u=3*u-1`then Total=total+u . We get 2 $+ 5 + 14 + 41 = 62$. The reverse order would successively add 2.5.14 to the total already initialized to 2 and omit 41.

Implementation 3. In row 0, 3 $\times 2^0 - 1 = 2 = a_0$. Fixing $n \in \mathbb{N}$ and suppose $a_n = 3 \times 2^n - 1$. Then $a_{n+1} = 2(3 \times 2^n - 1) + 1 = 3 \times 2^{n+1} - 1$. Initialization and inheritance make it possible to conclude that all $n \in \mathbb{N}$.

Implementation 4. $u_0 = 0$ gives the result null: inheritance is correct but never started. Add $u_0 \geq 2$, then check $P(0)$. Do not confuse the validity of an involvement with that of its hypothesis.

Application 5. In row 0: 1 $= 1^2$. Under the ranking hypothesis n, the sum at the rank $n + 1$ value $(n+1)^2 + 2(n+1) + 1 = (n+2)^2$. Conclude by induction. The new term has $n + 1$.

Implementation 6. Arithmetic Sequence $a_n = 40 + 5n$; geometric sequence $b_n = 40(1{,}1)^n$. First costs : 40.45.50 and 40.44.48.4 Cumulations: $10(40 + 85)/2 =$ EUR 625; 40 $\frac{1{,}1^{10}-1}{0{,}1} \simeq 637{,}50$ euros. The three the first few years are not sufficient to predict the order of the cumulative costs.

Pooling and differentiation

Comparing the tracks to two before displaying the formula. For the hole proof, accept a personal formulation provided that the integer is fixed and the hypothesis is identifiable. Then remove the primers. At S13, return to the initialization error with $u_0 = -1$. Observe the writing and not just the numerical result.

Conduct and correction

Benchmarks for the teacher

In S2, immediately distinguish the rank and value of the term. In S4, run a loop in the hand before Python. In S5 and S7, make verbalize "we assume in the rank n, we demonstrate at the level $n+1$". The sum and divisibility arrive in S10, after DS1: do not evaluate them at DS. In S11, the search for the fixed point is an algebraic technique; the theorem of For fragile students, provide the property $P(n)$ and let the three steps write; for the more advanced, remove the suggestion of auxiliary sequence.

Corrected activity

1. $u_0 = 100$ L. After one day, there is 0 left $,8u_0$ L, then add 30 L: $u_1 = 110$ L. Similarly, $u_2 = 0{,}8 \times 110 + 30 = 118$ L. For everything $n \geq 0$, $u_{n+1} = 0{,}8u_n + 30$.

2. Balance volume L should satisfy $L = 0{,}8L + 30$, either $L = 150$ L. Demontrons $P(n)$: « $100 \leq u_n < 150$". In row 0, $u_0 = 100 \in [100, 150[$. Fixing $n \geq 0$ and suppose $P(n)$ True. $0{,}8 > 0$,

$$110 = 0{,}8 \times 100 + 30 \leq 0{,}8u_n + 30 < 0{,}8 \times 150 + 30 = 150.$$

So 100 $\leq u_{n+1} < 150$ and $P(n+1)$ is true. By induction, the bounds is true for everything $n \in \mathbb{N}$.

3. $v_{n+1} = u_{n+1} - 150 = 0{,}8u_n - 120 = 0{,}8(u_n - 150) = 0{,}8v_n$. The future (v_n) is geometric, of common ratio 0.8 and first-term $v_0 = -50$. Thus $v_n = -50(0{,}8)^n$then $u_n = 150 - 50(0{,}8)^n$.

4.
```
def first_day():
    n, u = 0, 100
    while u <= 145:
        u = 0.8*u+30
        n += 1
    return n
```

The loop returns 11. Indeed, $u_{10} \simeq 144{,}63 < 145$ and $u_{11} \simeq 145{,}71 > 145$. In addition, $u_{n+1} - u_n = 10(0{,}8)^n > 0$: the threshold has not been crossed before. The calculation of these two values guarantees here the stop, without yet requiring a convergence theorem.

5. The simulation only computes a finite number of terms. the framework for all integers; it excludes any 150 L exceedance.

E1. $u_4 = 5 - 3 \times 4 = -7$. The sequence is arithmetical of common difference -3 and $u_{20} = -55$. The sum includes: 21 terms: $\sum_{k=0}^{20} u_k = \frac{21(u_0+u_{20})}{2} = \frac{21(5-55)}{2} = -525$.

E2. The sequence is geometric of common ratio 2 and first term $v_0 = 3$. Therefore $v_n = 3 \times 2^n$. The sum includes: $n+1$ terms and value $\frac{3(1-2^{n+1})}{1-2} = 3(2^{n+1} - 1)$.

E3. Poisons $P(n)$: « $3^n \geq 2n+1$". In row 0, $3^0 = 1 = 2 \times 0 + 1$. Fixes $n \geq 0$ and suppose $P(n)$ True. By multiplying by $3 > 0$, $3^{n+1} \geq 6n + 3$. Gold $6n + 3 - (2n+3) = 4n \geq 0$. Thus $3^{n+1} \geq 2(n+1) + 1$, what is $P(n+1)$. By induction, $P(n)$ is true for everything $n \in \mathbb{N}$.

E4. For $n \geq 0$, $u_n = 1 - \frac{1}{n+2}$. We have $0 < \frac{1}{n+2} \leq \frac{1}{2}$, therefore $\frac{1}{2} \leq u_n < 1$. In addition,

$$u_{n+1} - u_n = \frac{1}{n+2} - \frac{1}{n+3} = \frac{1}{(n+2)(n+3)} > 0.$$

The continuation is strictly increasing, reduced by $\frac{1}{2}$ plus 1.

E5. Note: $P(n)$: "There is an integer k such as $8^n - 1 = 7k$ For $n = 0, 8^0 - 1 = 0 = 7 \times 0 : P(0)$ It's true. $n \geq 0$ and suppose $P(n)$ True. So there is an integer k such as $8^n - 1 = 7k$. Then

$$8^{n+1} - 1 = 8(8^n - 1) + 7 = 8 \times 7k + 7 = 7(8k + 1).$$

Like 8 $k + 1$ is integer, $P(n + 1)$ It's true. By induction, 7 divides 8 $^n - 1$ for all $n \geq 0$.

Corrected from DM1

1. Poisons $P(n)$: « $2 \leq u_n < 6$". In row 0, $u_0 = 2$ check the frame. Fixes $n \geq 0$ and suppose $P(n)$ That's right. $\leq \frac{u_n}{2} + 3 < 6$; in particular 2 $\leq u_{n+1} < 6$. By induction, $P(n)$ is true to We're deducing from it. $u_{n+1} - u_n = 3 - \frac{u_n}{2} = \frac{6-u_n}{2} > 0$. The continuation is strictly increasing.

2. Equation $L = \frac{L}{2} + 3$ gives $L = 6$. Pose $v_n = u_n - 6$. Then $v_{n+1} = \frac{u_n}{2} - 3 = \frac{v_n}{2}$ and $v_0 = -4$. So $v_n = -\frac{4}{2^n}$ and $u_n = 6 - \frac{4}{2^n}$.

3. By sum linearity and geometric formula,

$$\begin{aligned} S_n &= 6(n+1) - 4\sum_{k=0}^{n}(\frac{1}{2})^k \\ &= 6(n+1) - 8(1 - 2^{-(n+1)}). \end{aligned}$$

4. $u_n > 5{,}99$ is equivalent to $\frac{4}{2^n} < 0{,}01$, or 2 $^n > 400$. As 2 $^8 = 256 \leq 400 < 512 = 2^9$ and (2^n) is increasing, the smallest row is 9.

5. The average is less than 6 and the difference is 6

$$6 - \frac{S_n}{n+1} = \frac{8(1 - 2^{-(n+1)})}{n+1}, \qquad 0 < 6 - \frac{S_n}{n+1} < \frac{8}{n+1}.$$

To make this difference less than precision $\varepsilon > 0$, just take $n + 1 > \frac{8}{\varepsilon}$. It's a quantitative control of the deviation, which does not yet require a theorem on the limits.

Corrected work in the manual or an anal

Fiscal year 22. The four translations are, for all $n \in \mathbb{N} : u_{n+1} \geq u_n$; $v_n \geq 5$; $w_n \leq -1$; $t_n > 0$. The monotonicity compares two consecutive terms; a limit compares each term to the same number.

Exercise 25. We have $u_0 = -5$ and $u_{n+1} = \frac{2}{3}u_n - 1$.

Initialization. $u_0 = -5 \leq -3$: the property is true in row 0.

Heredity. Fixes $n \in \mathbb{N}$ and suppose $u_n \leq -3$. Multiplication by $\frac{2}{3} > 0$ keeps the order, so $\frac{2}{3}u_n \leq -2$. Thus $u_{n+1} = \frac{2}{3}u_n - 1 \leq -3$.

Conclusion. By induction, $u_n \leq -3$ for all $n \in \mathbb{N}$. Then, $u_{n+1} - u_n = \frac{2}{3}u_n - 1 - u_n = -\frac{1}{3}u_n - 1$. The previous bound gives $-\frac{1}{3}u_n \geq 1$, where $u_{n+1} - u_n \geq 0$. The continuation is increasing. sense of inequality here comes from the negative factor $-\frac{1}{3}$.